



# Parks on the Air Activator Course

Session 1 — The POTA Universe

TavaOne Education · Pinellas STEM Radio Lab · W4GGJ

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# Today's Session

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- What is POTA — history, roles, the community
- Park references and pota.app database
- Setting up your pota.app account
- What makes a valid activation
- Park logistics and Leave No Trace
- **Activity: Find your first 3 parks**



# What is POTA?

Program history, roles, and the community



# Parks on the Air — Overview

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- Started as World Wide Flora & Fauna (WWFF), a global program
- US-specific POTA program launched 2017 as K-POTA
- Became fully independent in 2020 — now largest portable ops program worldwide
- Over 50,000 parks in 100+ entities
- Completely FREE — no membership fees, no pay walls
  
- US park types:
  - National Parks, Monuments, Recreation Areas
  - National Forests and Grasslands
  - National Wildlife Refuges, Army Corps sites
  - State Parks, State Forests, State Recreation Areas
  - Many county parks (check the database)



# Activator vs. Hunter

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## Activator

- Goes to a POTA park
- Sets up a portable station
- Calls CQ POTA on the air
- Needs 10 QSOs for a valid activation
- Uploads log to pota.app



## Hunter

- Operates from home or anywhere outside the park
- Monitors spots on pota.app
- Calls the activator when spotted
- Gets credit for each park worked
- Chases rare parks for awards



## Park-to-Park (P2P)

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- When two activators work each other, it's a Park-to-Park contact
- Both get regular activation/hunter credits
- Both also get P2P credit — tracked separately on [pota.app](#)
  
- How to find P2P contacts:
  - Watch the spots page — look for other activators on the air
  - Call CQ and a fellow activator may respond
  - Listen for "de [callsign] at [park ref]" in their response
  
- P2P contacts count toward your 10 QSO minimum
- POTA awards include P2P-specific milestones



# Park References & `pota.app`

Finding parks, reading park pages, using the database



# The K-Reference System

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- Every park has a unique reference number
- US parks: K-XXXX (4 digits) — example: K-2119
- Some parks also have a KFF- number (old WWFF format — same park)
- Other entity prefixes:
  - VE- = Canada
  - DL- = Germany
  - VK- = Australia
  - (over 100 entities worldwide)
- You need the correct K-reference BEFORE you activate
- Look it up on [pota.app](https://pota.app) — park name search or map view
- Put the reference in your log as SIG\_INFO field (we cover this in Session 3)



## Reading a Park Page on pota.app

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- Every park has a dedicated page — tap any pin on the map
- Key information on a park page:
  - Reference (K-number) — copy this exactly
  - Park name and managing entity
  - Total lifetime activations
  - Total QSOs logged from this park
  - Most recent activation date and activator
  - Leaderboard — who has activated it most
  - Hunter log — who has worked this park
- A park with 0 activations is called "unactivated"
- Hunters love unactivated parks — rare credit for them



# Your **pota.app** Account

Setup, the activator dashboard, and HamAlert



# Creating Your Account

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- Go to [pota.app](https://pota.app) — hit Register
- Use your callsign as your username (must match your FCC license)
- Verify email
  
- Profile setup:
  - Profile → Edit Profile
  - Enter callsign, name, QRZ link
  - Optionally link LoTW certificate
  
-  Callsign must match exactly
- Hunters log your callsign when they work you
- System matches it to your account automatically
- Mismatch = hunters don't get credit = unhappy hunters



# Your Activator Dashboard

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- Shows after you log in:
  - Total activations (valid count)
  - Total parks activated
  - Award progress — badges at 1, 10, 25, 50, 100, 250, 500, 1000
  - Recent activation history
  - Pending activations (uploaded, processing)
- HamAlert ([hamalert.org](https://hamalert.org)):
  - Free push notification service for hunters
  - Set up alerts for specific parks you want to catch
  - Also useful to monitor if your own spot went live
  - Can text you when a specific callsign is spotted



# Valid Activation Rules

What counts, what doesn't, and the UTC clock



# The Five Rules

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- 1. 10 QSOs minimum to earn activator credit
  - 2. All QSOs from within park boundaries
  - 3. Same UTC calendar day for all contacts
  - 4. Your callsign as control operator
  - 5. Valid ADIF log uploaded to [pota.app](https://pota.app)
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- If you make fewer than 10 QSOs:
    - Upload the log anyway
    - You don't get activator credit
    - Hunters DO get credit — they earned it
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- Overlapping parks: count QSOs toward both simultaneously



# UTC — The Activator's Time Zone

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- All POTA logs use UTC — not local time
- Florida time conversions:
  - EDT (summer) = UTC – 4 hours
  - EST (winter) = UTC – 5 hours
- Example: 7:30 PM EDT = 23:30 UTC (same date)
- Example: 9:00 PM EDT = 01:00 UTC (NEXT date)
-  Late activations: watch the UTC midnight rollover
- If QSOs span two UTC dates, log the split carefully
- HAMRS handles this automatically — trust the app
- Set your phone or radio clock to UTC before activating



# Modes That Count vs. Don't Count

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## ✓ Valid POTA Modes

- SSB (phone) — most common
- CW — short exchanges, very popular
- FT8 / FT4 — highest rate per hour
- FM — VHF/UHF, 147.555 simplex
- RTTY, PSK31, other digital RF modes
- Any RF mode in the rulebook

## ✗ Does Not Count

- Echolink (internet-linked)
- AllStar, IRLP, D-STAR repeater internet
- DMR via internet gateway
- Winlink HF is OK; Winlink via internet is not
- Any connection where RF is not end-to-end



# Park Logistics & Leave No Trace

Research before you go, etiquette, safety



## Before You Go — Research Checklist

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- Park hours — many close at sunset; check the website
- Entry fees — budget \$5–\$10 for most state parks
- Permit requirements — some NPS parks require one for antenna setups
- Cell coverage — you need a signal to self-spot; check coverage maps
- Parking — is there a spot close to a suitable operating area?
  
- Have a backup plan:
  - If the park is closed, have a nearby alternative K-reference ready
  - If weather turns, know if there's a covered pavilion
  
- Tip: [pota.app](https://pota.app) past activation comments often mention logistics details other activators noted



# Antenna Placement and Leave No Trace

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- Throwing wire into trees: generally fine with a soft throw bag
- Telescoping poles (squid poles): excellent for parks — no tree needed
- Driving ground stakes: fine in camping/picnic areas
- Never leave attached hardware — remove everything when you leave
  
- Leave No Trace in practice:
  - Pack everything in, pack everything out
  - Use existing paths and clearings for setup
  - Loop wire on trees — don't hard-tie (trees move in wind)
  - Leave tables cleaner than you found them
  - Position generators away from other visitors
  
- If a ranger approaches: be friendly, explain POTA, show your license



# Safety for Solo Activations

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- Tell someone where you're going and when you'll be back
- Keep your phone charged — it's your lifeline
- Know the nearest hospital or urgent care
- Florida-specific hazards:
  - Lightning — most dangerous RF hazard outdoors
  - When thunder starts: antenna down immediately
  - A long wire in a storm is a lightning rod
  - Head to your car — it's a Faraday cage
  - Heat — hydrate, know the signs of heat exhaustion
  - Wildlife — alligators near water, snakes in vegetation
- If you feel unsafe: pack up and leave. No activation is worth it.



## KNOWLEDGE CHECK

**How many QSOs do you need for a valid POTA activation?**

A. 5 QSOs

B. 10 QSOs

C. 15 QSOs



## KNOWLEDGE CHECK

**You make 8 QSOs before your battery dies. What should you do?**

A. Don't upload — it's not a valid activation

B. Upload the log anyway so hunters get credit

C. Go home and pretend it didn't happen



## KNOWLEDGE CHECK

**An Echolink contact made from inside a National Forest — does it count for POTA?**

A. Yes — you're inside a park, so it counts

B. Only if the other station is also in a park

C. No — Echolink uses the internet; POTA requires RF end-to-end

# Find Your First Parks

- Open [pota.app](https://pota.app) → map view → navigate to your home area
- Find 3 POTA parks within 30 miles
- For each: note the K-reference, name, total activations, last activated date
- Pick your favorite as a target for your first solo activation
- **Share your pick with the group — and why**



## Session 2 Preview

# Field Station Fundamentals

- Radio selection for POTA — IC-705, FT-891, Xiegu, KX2/KX3
  - Antennas — EFHW, wire dipoles, linked dipoles, verticals
  - Power options — LiFePO4, solar, capacity math
  - Building your go-kit — weight, checklist, pack-and-go
  - **Hands-on: set up the club IC-705 + EFHW on the patio**
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